

BUSINESS INSIGHTS & RECOMMENDATIONS REPORT

Sample Superstore — Advanced Analysis & BI Dashboard

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Executive Summary

This report extends the Week 2 Sample Superstore analysis with deeper, evidence-based investigation into profitability, discounting, and the post-2016 sales pattern. Six new KPIs and seven DAX measures were added, alongside three new dashboard pages built around specific business problems. The new Profit Growth % measure confirms and quantifies the decline Week 2 first flagged: profit fell 47.18% from 2016 to 2017. Three concrete, addressable issues were identified and are detailed below, and the evidence points to Furniture as the primary driver of that decline.

Updated KPI Summary

The table below combines the five KPIs carried over from Week 2 with six new measures developed in Week 3.

Metric	Week 2 Value	Week 3 Addition	Value
Total Sales	\$307,800.33	—	—
Total Profit	\$24,962.70	—	—
Profit Margin	8.11%	—	—
Total Orders	1,334	—	—
Average Sale	\$230.73	—	—
Total Customers	—	New	644
Loss-Making Orders	—	New	243 (18.2% of all orders)
Total Discount Given	—	New	\$204.49
Discount % of Sales	—	New	0.07%
Profit Growth (YoY)	—	New	-47.18%

Key Business Insights

Insight 1: Profit Declined 47.18% From 2016 to 2017, Confirming the Post-2016 Concern

The new Profit Growth % measure quantifies what Week 2 first flagged as a concerning trend: profit fell 47.18% between 2016 and 2017, a steeper drop than the sales figures alone suggested. This is not a company-wide erosion, however — as Insight 2 shows, it is concentrated in one category.

Insight 2: The Decline Was a Furniture Problem, Not a Company-Wide One

Breaking Sales by Year down by Category shows Furniture sales fell sharply from roughly \$30K in 2016 to \$20K in 2017, while Technology grew over the same period (from about \$34K to over \$30K, remaining the largest category) and Office Supplies stayed essentially flat. Furniture's decline explains most of the 47.18% profit drop identified in Insight 1 — Technology's growth partially offset it, but not enough to prevent an overall fall.

Insight 3: 18.2% of All Orders Actively Lose Money

Of 1,334 total orders, 243 (18.2%) generated a loss. This is a meaningfully large share of transactions destroying value rather than creating it, and helps explain why the overall profit margin remains thin despite healthy total sales.

Insight 4: A Small Number of Products Account for Outsized Losses

The single largest loss-making product, the Cubify CubeX 3D Printer, lost \$3,839.99 on \$7,999.98 of sales — nearly a 48% loss rate on that item alone. Of the seventeen biggest loss-makers identified, twelve are Furniture products (bookcases, tables, chairs), directly connecting the loss-making-products investigation to the Furniture profit gap flagged in Week 2.

Insight 5: Discounting Is Minimal and Does Not Explain the Furniture Losses

Total discounting across the dataset is remarkably small: discounts represent only 0.07% of total sales value (\$204.49 against \$307,800 in sales). The discount-versus-profit scatter chart shows the highest-profit products cluster at low discount levels, and the biggest losses (like the 3D printer) are not explained by heavy discounting. This rules out discounting as the primary driver of Furniture's weak margins and points instead toward cost structure — likely shipping, supplier pricing, or per-unit cost issues on bulky items.

Insight 6: Profit Follows a Seasonal Pattern Peaking Mid-Year

The monthly profit trend shows a rise into a peak around June–July, followed by a decline through late summer and a choppy recovery into Q4. This seasonal shape is useful for planning inventory, staffing, and promotional timing around the mid-year peak rather than assuming a flat, year-round demand curve.

Business Recommendations

Recommendation 1: Investigate Furniture's Cost Structure, Not Its Discounting

Action: Conduct a supplier and shipping cost review specifically for Furniture, especially bookcases, tables, and chairs. Since discounting is minimal, the loss driver is more likely per-unit cost (freight, packaging, supplier pricing) than promotional pricing.

Why: Twelve of the seventeen biggest loss-making products are Furniture items, and the discount data rules out heavy discounting as the cause — cost structure is the remaining explanation.

Expected Impact: Correcting even a few of the largest Furniture cost drivers could meaningfully improve the category's contribution to the company's still-thin 8.11% overall margin.

Priority: HIGH

Recommendation 2: Review or Discontinue the Worst Individual Loss-Makers

Action: Conduct a product-level pricing and viability review for the top loss-making products, starting with the Cubify CubeX 3D Printer (-\$3,839.99) and the Riverside Palais Royal Lawyers Bookcase (-\$1,665.05). Either reprice these items to cover cost, or discontinue them if margins cannot be corrected.

Why: A small number of products are responsible for a disproportionate share of total losses; fixing or removing them has an outsized effect relative to the effort required.

Expected Impact: Removing or repricing the top five loss-makers alone would recover roughly \$8,000 in lost profit — about a third of current total profit.

Priority: HIGH

Recommendation 3: Investigate the Root Cause of the 18.2% Loss-Making Order Rate

Action: Segment the 243 loss-making orders by category, region, and discount level to identify whether losses cluster around specific shipping methods, order sizes, or customer segments, beyond what the category-level view already shows.

Why: Nearly one in five orders currently loses money; understanding the pattern behind these orders (not just which products) will surface operational fixes beyond product-level pricing.

Expected Impact: Reducing the loss-making order rate even modestly would directly improve overall profit margin without requiring any increase in sales volume.

Priority: MEDIUM

Recommendation 4: Plan Inventory and Promotions Around the Mid-Year Profit Peak

Action: Use the monthly profit trend to align inventory build-up, staffing, and promotional campaigns with the June–July peak period, and investigate what drives the late-summer dip.

Why: The seasonal pattern is now visible in the data and was not part of the Week 2 analysis; planning around it is a low-cost way to capture more of the existing seasonal demand.

Expected Impact: Better alignment of inventory and marketing spend with the actual seasonal curve should improve both sales capture and cost efficiency during peak months.

Priority: MEDIUM

Recommendation 5: Continue Prior Recommendations: Expand Technology, Build Consumer Loyalty, Adopt the Dashboard as a Standing Tool

Action: Carry forward the three lower-risk Week 2 recommendations: continue investing in Technology (now confirmed as the category driving growth despite the overall 2016–2017 dip), build a Consumer segment loyalty program, and formally adopt this dashboard as a recurring tool in leadership meetings, now expanded with Week 3's deeper analysis pages.

Why: Week 3 analysis reinforces rather than contradicts these three original recommendations — Technology's growth is now directly evidenced, and the expanded dashboard is a stronger tool than the Week 2 version.

Expected Impact: These initiatives compound the gains from Recommendations 1-4 by growing the healthy part of the business while the problem areas are being fixed.

Priority: MEDIUM

Top Loss-Making Products (Supporting Detail)

The table below lists the five largest individual loss-makers identified in the Week 3 analysis, referenced in Insight 4 and Recommendation 2.

Product	Category	Sales	Profit
Cubify CubeX 3D Printer Triple Head Print	Technology	\$7,999.98	-\$3,839.99
Riverside Palais Royal Lawyers Bookcase	Furniture	\$3,083.43	-\$1,665.05
Chromcraft Bull-Nose Wood Oval Conference Tables & Bases	Furniture	\$5,950.58	-\$1,630.90
Hoover Upright Vacuum With Dirt Cup	Office Supplies	\$463.25	-\$1,181.28
Chromcraft 48" x 96" Racetrack Double Pedestal Table	Furniture	\$1,667.33	-\$833.66

Conclusion

The Week 3 analysis confirms and sharpens the concern Week 2 first raised: profit fell 47.18% from 2016 to 2017, and discounting is not the cause. The core weaknesses are narrow and addressable: a Furniture cost-structure and sales-decline problem, a small set of individually loss-making products, and an 18.2% loss-making order rate. Because these issues are concentrated in Furniture rather than spread company-wide, targeted action on Furniture costs and the worst-performing products offers a clear, high-leverage path to recovering profit — especially since Technology's continued growth shows the underlying business model still works.